



Pak-Austria Fachhochschule: Institute of Applied Sciences and Technology

Synopsis for the Degree of Master of Science (MS)

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Program	MS Artificial Intelligence		
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Title of Research Proposal	CREAIS: Cognitive Requirement Engineering for the Next Generation of Ethical Artificial Intelligence Systems		
Summary of the Research Proposal Artificial intelligence (AI) technologies are increasingly used in sensitive areas of criminal justice, healthcare, banking, and self-driving cars. Despite the numerous benefits of such technologies, it also raises some grave ethical concerns including discrimination, discrimination, lack of transparency, threat of privacy, and lack of responsibility. Although functional and quality requirements are the primary concerns of the traditional requirements engineering (RE)			

methodologies, ethical concerns are hardly ever consistently integrated in system development. It is due to that fact that such moral principles as justice, explicability, and reliability often remain abstract and hard to implement. To overcome this issue, it is required to develop a specifically designed AI-based RE framework that would have both incorporated ethical considerations at an early stage of system design and ensured that ethical standards would be verified through the development lifecycle.

The proposed research offers a larger RE framework to AI systems through the addition of ethical considerations to elicitation, analysis, specification, validation, and traceability stages of the process. It will be possible to convert the ethical principles of fairness, responsibility, privacy, and explicability into how quantifiable parameters and be able to validate them using benchmark datasets (including UCI Adult Income, COMPAS, and German Credit) and interpretable models (including logistic regression and decision trees). A web-based application will be developed to record the ethical requirements, compare them using fairness and explainability metrics, and maintain a trace between the datasets, models and the evaluation outcomes. The expected input of this project is a helpful framework and validation platform that enhances the openness, responsibility, and moral soundness of AI development.

Introduction

The use of artificial intelligence (AI) technologies in such a field as health care, banking, criminal justice, and self-driving cars is becoming increasingly frequent. Such systems are actively used to support or replace the human decision-making in the field of employment where the social and ethical consequences are pressing. Efficiency and cost-saving, scalability are some of their potential advantages, but the issues of bias, unfairness, invasion of privacy, lack of transparency, and accountability are very critical [1], [2]. Unregulated AI can enhance discrimination and lack of trust, which is manifested in the popular cases of bias in recidivism prediction and credit score systems [3]. To mitigate such issues, the ethical issues must be addressed as a regular aspect during the construction of AI systems rather than being viewed as an accidental factor.

Decision of what a system should do and with what constraints is one of the significant tasks of requirement engineering or RE. RE has conventionally paid more attention to its functional requirements i.e. capabilities of the system, i.e. quality criteria i.e. performance, usability, and security. Nonetheless, as soon as AI systems start influencing individuals and society, in general, RE will be forced to be modified in a manner that would accommodate the ethical standards like

responsibility, privacy, explainability, and fairness [1], [4]. The fact that the ideas of ethical practices need to be transformed into more quantifiable, visible and traceable practices is a major issue because such ideas have the tendency to be abstract and circumstantial. The gap between the ethical ideals on the high level and its implementation in the creation of AI is explained by the fact that the modern processes and technologies of RE do not tend to give systematization in addressing these issues [2], [5].

It has a lot of research studies, which have identified this problem as well as given some partial solutions. It has been observed that despite all the engagements, not all the ethical issues such as openness and fairness are always incorporated into the processes of RE [1]. According to some researchers, ethical requirements of AI systems ought to be considered similarly to the legal requirements that are based on the codes and principles [2]. In a bid to formalise and certify ethical requirements, assurance-case-based models have been proposed [7] and ontology-based modelling [6]. The explainability need has as well been examined using the user-centered ways [8], [9]. Cumulatively these studies demonstrate the significance of the ethics consideration in the process of RE, however, there is a lack of lifecycle traceability, tool support and quantifiable considerations.

Our research fills these gaps by publishing AI-specific RE framework that will systematically consider the ethical considerations in the lifecycle of the development process. The framework will also incorporate the ideas of ethics as being part of the first-class requirements into the normal RE processes of elicitation, analysis, specification, validation, and traceability. With the help of structured templates and benchmark data like UCI Adult Income, COMPAS, and German Credit, even such abstract notions as fairness, privacy, accountability and explicability will be transformed into quantifiable requirements. The compliance with ethical standards will be checked with the help of interpretable models (Decision Trees and Logistic Regression) and explainability tools (SHAP and LIME). A web based validation tool will also be designed to record, test and trace requirements throughout the lifetime as a way of establishing accountability and constant monitoring. The study will seek to fill the gap between theoretical ethical principles and the real world of AI engineering through the approach and act as a practical framework and tool.

Problem Statement

The current Requirement Engineering (RE) methods of Artificial Intelligence (AI) systems would not adequately characterize, operationalize, and validate ethical needs such as accountability, privacy, explainability, and fairness. The AI systems pose a threat to bias, obscurity, and lack of credibility since ethical standards are yet to be concrete and have no quantifiable parameters, tool support, and their lifecycle traceability.

Purpose of the Study

The objective of this project is to design and introduce an AI-based Requirement Engineering (RE) framework that systematically incorporates such ethical requirements as responsibility, privacy, explainability, and fairness into all phases of AI system creation. To achieve this, quantifiable requirement templates will be established, operationalized with benchmark datasets and models, and a web based validation tool that ensures compliance to ethical standards, continuous testing and traceability will be generated.

Objectives

1. To develop a Requirement Engineering (RE) framework that applies to AI which introduces the ethical limitations of privacy, responsibility, explainability, and fairness into the process of elicitation, analysis, specification, validation, and traceability in the development of AI systems.
2. To develop and implement a web-based validating tool based on the interpretable AI models and benchmark datasets that enables the recording, measure and continuous monitoring of ethical standards.

Research Questions:

1. How do we incorporate the explainability, privacy, fairness, and accountability in the AI requirements?
2. How can an online tool be applied to monitor and observe ethical standards in the creation of AI?

Research Methodology

Research methodology provides procedural steps and specified plan which will guide this research to achieve its objectives. Since the objective of the given research is to make Requirement Engineering (RE) accommodate ethical requirements to the AI systems, tool development, data-driven verification, and a framework of finding development will be applied.

1. Review of Literature and Analysis of Gaps

The first step is to review existing Requirement Engineering (RE) frameworks, the Responsible AI principles, and ethical requirement models in detail. This assessment will reveal limitations of current procedures, in particular, with respect to integration of ethical concepts such as accountability, privacy, explainability, and fairness. A gap analysis will be used to demonstrate where the existing solutions are inadequate to operationalize these principles into tangible needs to demonstrate the need of a new framework.

2. Development of Frameworks

Depending on the findings of the literature research, an AI-specific RE framework will be developed. Ethical ideals will be defined by structured requirement templates that transform intangible ideas to quantifiable and testable components. These will be mapped to functional and quality requirements in order to ensure that ethics are systematically integrated to the elicitation, analysis, specification, validation, and traceability processes of the RE process. The framework will also align life cycle management processes with ethical standards to ensure that they remain sustainable during the evolution of the AI systems.

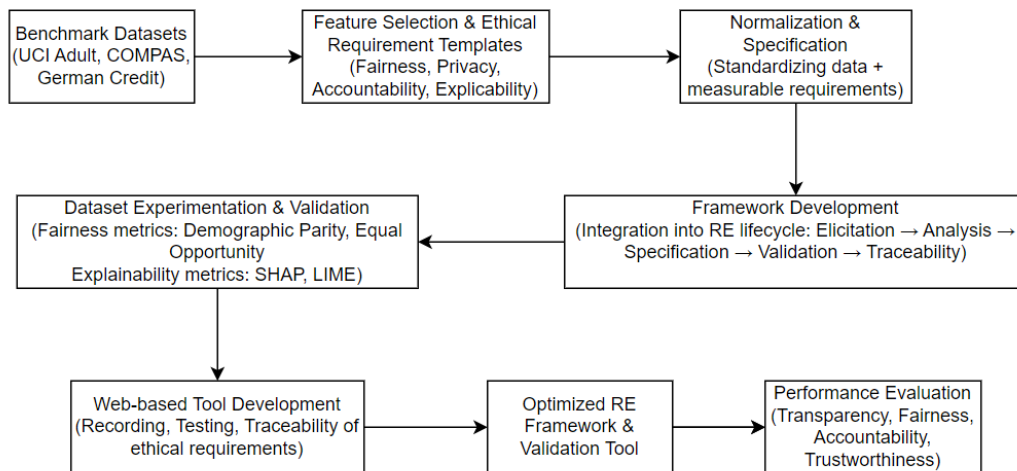
3. Validation and Experimentation with Datasets

The operationalization of ethical standards will be based on benchmark datasets, including German Credit, Compas and UCI Adult Income. Such interpretable models like decision trees and logistic regression will be served as test cases and this will help in making the results clear and understandable. The compliance with the ethical standards will be achieved by using

explainability methods (e.g., SHAP, LIME) and fairness measures (e.g., demographic parity, equal opportunity). This action will show how the framework will turn the theoretical ideas into the practical deliverables of the real-life AI projects..

4. Tool Creation and Application

A web-based validation tool will be developed to ease in the process of recording, testing and tracking of ethical criteria. Technology will also introduce some explainability and fairness features, which will enable AI models to be supervised constantly regarding the morals. It will also maintain traceability by relating requirements with information, models and evaluation results. The utility and effectiveness of the tool will be evaluated by the case studies that will also provide insights into how the tool will contribute to bridging the chasm between the theoretical models of RE and the development of the feasible AI systems.



Ac

Examples (as applicable)

Objective 1: Methodology

- The stages of AI system development elicitation, analysis, specification, validation, and trace ability are complemented with the stage of ethical requirements in order to come up with an Requirement Engineering (RE) model, which will also be AI-oriented.

- To create loopholes, look at the literature of the existing RE frameworks, Responsible AI guidelines and models of ethical requirements.
- Definitely set ethical guidelines (explainability, privacy, responsibility, and fairness) with the help of systematic templates.
- Transform the moral values into measurable values based on interpretable models(Logistic Regression, Decision Trees) and benchmark data(UCI Adult Income, COMPAS, German Credit).
- These are requirements, which have to be considered during the formulation of the extended RE model, in his lifecycle phases.

Objective 2: Methodology

- To design and deploy an online validation device and rely on interpretable AI models and benchmark data that would allow recording, measuring, and continuously track ethical standards.
- Design a web application that is traceable, has validation and requirement capturing modules.
- The tool contains fairness and explainability, such as equal opportunity, demographic parity, SHAP, and LIME.
- Benchmark Test the tool using benchmark datasets and compare model results.
- Show how the ethical principles may be tracked and verified in the course of the AI development in order to justify the technology.

Literature Review

The study in [1] identifies critical ethical issues in AI deployment across healthcare, banking, and transportation sectors, highlighting algorithmic bias as a persistent challenge. It emphasizes how bias emerges from training data and design choices, leading to discriminatory outcomes. While comprehensive in identifying problems, the study lacks concrete mitigation frameworks. Future work focuses on developing standardized bias assessment protocols.

The study in [2] examines practical barriers to implementing ethical AI in organizational contexts, documenting how transparency deficits undermine accountability. Through case studies in financial and healthcare AI, it reveals disconnects between ethical policies and

development practices. The research calls for integrated governance frameworks but offers limited technical implementation guidance. Future work targets bridging policy-practice gaps.

The study in [3] proposes "datasheets for datasets" as standardized documentation templates to enhance AI transparency. Each datasheet captures dataset characteristics, collection methods, recommended uses, and limitations. Evaluated across research and industry datasets, the approach improves reproducibility and bias awareness. Limitations include voluntary adoption and documentation maintenance challenges. Future work focuses on automated datasheet generation.

The study in [4] introduces "model cards" as standardized reporting frameworks for AI models, detailing performance across demographic groups, intended uses, and ethical considerations. Tested with commercial ML models, model cards improve stakeholder understanding and facilitate fairness assessments. Challenges include quantifying complex model behaviors and maintaining updated documentation. Future work explores dynamic model cards with real-time metrics.

The study in [5] presents the NIST AI Risk Management Framework (AI RMF 1.0), emphasizing lifecycle integration of ethical considerations from design through deployment. The framework provides structured processes for risk identification, assessment, and mitigation. While comprehensive, implementation requires significant organizational commitment. Future work focuses on sector-specific adaptations and lightweight versions for SMEs.

The study in [6] describes the IEEE 7000-2021 standard for addressing ethical concerns during system design, providing a model process with defined stages and artifacts. The standard facilitates traceability from ethical principles to technical requirements. Validation in healthcare AI projects shows improved ethical compliance but increased documentation overhead. Future work targets toolchain integration and certification processes.

The study in [7] develops comprehensive taxonomies for Explainable AI (XAI) methods, categorizing techniques by explanation type, scope, and technical approach. The taxonomy facilitates method selection based on application requirements and stakeholder needs. While theoretically robust, practical application requires significant expertise. Future work focuses on decision-support tools for XAI method selection.

The study in [8] proposes SHAP (SHapley Additive exPlanations) as a unified framework for interpreting model predictions using cooperative game theory. The method provides both global

feature importance and local instance explanations. Evaluated across diverse ML models, SHAP demonstrates superior consistency compared to alternative methods. Computational complexity remains a limitation for large-scale deployment. Future work optimizes approximation algorithms.

The study in [9] introduces LIME (Local Interpretable Model-agnostic Explanations) for explaining individual predictions by approximating complex models locally with interpretable surrogates. The approach works with any classifier without model internals access. While flexible, explanations can be unstable with different random samples. Future work enhances stability and provides confidence estimates for explanations.

The study in [10] advocates for inherently interpretable models over post-hoc explanations in critical applications like healthcare and criminal justice. It compares interpretability-accuracy tradeoffs across model classes, finding interpretable models often suffice for many high-stakes tasks. The approach reduces explanation complexity but may sacrifice performance in complex domains. Future work develops hybrid interpretable-deep architectures.

The study in [11] formally defines "equality of opportunity" as a fairness metric requiring equal true positive rates across demographic groups. The mathematical formulation enables precise measurement and optimization. Applied to hiring and lending datasets, the metric reveals discrimination patterns invisible to simpler fairness measures. Implementation challenges include reliable protected attribute collection. Future work explores proxy-resistant formulations.

The study in [12] examines demographic parity as a fairness criterion requiring equal selection rates across groups. It analyzes implementation approaches including pre-processing, in-processing, and post-processing techniques. While conceptually straightforward, demographic parity can conflict with meritocratic principles in competitive contexts. Future work develops context-aware fairness criteria balancing multiple ethical dimensions.

The study in [13] analyzes the UCI Adult Income dataset as a benchmark for fairness algorithm evaluation, documenting its characteristics, limitations, and appropriate use cases. The dataset enables standardized comparison of fairness interventions but contains historical biases reflecting

societal inequities. Researchers must account for these limitations when drawing conclusions. Future work develops more representative benchmark datasets.

The study in [14] investigates the COMPAS recidivism risk assessment dataset, revealing significant racial disparities in prediction accuracy and false positive rates. The analysis sparked widespread debate about algorithmic fairness in criminal justice. Dataset limitations include incomplete feature representation and ambiguous ground truth. Future work focuses on improved criminal justice risk assessment methodologies.

The study in [15] documents how even widely-used benchmark datasets like German Credit contain inherent biases that perpetuate discrimination if not properly addressed. It analyzes bias sources including sampling methods, labeling processes, and historical inequities. The research provides guidelines for bias-aware dataset usage but notes complete debiasing may be impossible. Future work develops bias quantification standards.

The study in [16] demonstrates how historical biases persist in AI systems despite technical fairness interventions, emphasizing the need for socio-technical solutions combining algorithmic and organizational changes. Case studies show limited impact of technical fixes alone. The research calls for holistic fairness approaches addressing power structures and decision contexts. Future work develops integrated socio-technical frameworks.

The study in [17] argues for defining explainability as a quantifiable non-functional requirement within RE processes, proposing metrics for different explanation aspects including completeness, fidelity, and comprehensibility. The approach enables systematic specification and validation of explanation requirements. Practical implementation requires stakeholder education and tool support. Future work develops standardized explanation quality metrics.

The study in [18] proposes using formal ontologies to structurally represent ethical requirements and relationships between ethical concepts in AI systems. The ontology enables reasoning about requirement consistency and completeness. While theoretically rigorous, ontology development requires significant expertise. Future work focuses on reusable ethical requirement ontologies for common domains.

The study in [19] introduces the E-LENS framework using assurance cases to link ethical requirements with evidence and validation stages. The approach provides structured argumentation for ethical compliance but can become complex for large systems. Future work develops tool support for assurance case management and automated evidence collection.

The study in [20] presents capability-driven approaches to translate high-level Responsible AI principles into measurable system capabilities through capability modeling and refinement. The method bridges abstract principles and concrete implementations but requires careful capability decomposition. Future work develops capability patterns for common ethical requirements.

The study in [21] documents challenges practitioners face in capturing, measuring, and validating ethical requirements through interviews with 45 AI developers. Key barriers include ambiguous ethical concepts, lack of metrics, and organizational resistance. The research identifies specific pain points but offers limited solutions. Future work develops practitioner-centered tools and processes.

The study in [22] conducts a systematic mapping of Requirements Engineering for AI (RE4AI) literature, identifying significant gaps in lifecycle integration of ethical requirements. The mapping reveals disproportionate focus on technical requirements over ethical considerations. The research provides a foundation for targeted research but doesn't address identified gaps. Future work focuses on under-researched ethical RE areas.

The study in [23] identifies traceability as a major deficiency in ethical RE approaches, with most methods failing to maintain reliable links between ethical principles and implementation artifacts. Without traceability, ethical compliance cannot be systematically verified. The research highlights the problem but offers limited solutions. Future work develops ethical requirement traceability frameworks.

The study in [24] examines validation approaches for ethical requirements, finding most methods lack automated tool support and rely on manual review. This limits scalability and consistency in

ethical validation. The research catalogs available validation techniques but notes significant integration challenges. Future work develops integrated validation toolchains.

The study in [25] presents a comprehensive roadmap for Responsible AI development emphasizing governance structures, accountability mechanisms, and lifecycle integration. The roadmap identifies maturity stages and transition paths but provides limited implementation details. Future work develops sector-specific roadmaps with concrete milestones.

The study in [26] describes the Software Engineering for Machine Learning (SE4ML) initiative focusing on adapting traditional SE practices for ML systems. It identifies unique challenges in testing, debugging, and maintaining ML systems. The research synthesizes best practices but notes limited empirical validation. Future work conducts longitudinal studies of SE4ML adoption.

The study in [27] introduces Microsoft's Fairlearn toolkit for assessing and improving AI fairness across demographic groups. The toolkit provides metrics, visualizations, and mitigation algorithms with scikit-learn integration. While comprehensive, integration into existing workflows requires adaptation. Future work expands mitigation techniques and improves scalability.

The study in [28] presents IBM's AI Fairness 360 (AIF360) toolkit with extensive fairness metrics and algorithms across the ML pipeline. The toolkit supports multiple fairness definitions and provides benchmarking capabilities. Complexity and computational requirements can hinder adoption. Future work simplifies APIs and optimizes performance.

The study in [29] examines technical and organizational barriers to integrating ethics tools like Fairlearn and AIF360 into development workflows. Key challenges include tool complexity, skill gaps, and process misalignment. The research identifies integration patterns but notes limited adoption data. Future work develops lightweight integration frameworks.

The study in [30] argues for integrating ethics validation at the requirement level rather than as post-hoc analysis, proposing architecture patterns for requirement-tool integration. This enables early validation but requires rethinking development processes. Future work develops reference architecture and integration guidelines.

The study in [31] emphasizes user-centric approaches in XAI where explanations are tailored to different stakeholder needs and cognitive styles. User studies reveal significant variation in explanation preferences across roles. The research provides design principles but lacks implementation frameworks. Future work develops adaptive explanation systems.

The study in [32] investigates how different XAI approaches affect user trust and decision-making in human-AI collaboration. Controlled experiments show explanation characteristics significantly influence appropriate trust calibration. The research provides empirical evidence but limited design guidance. Future work develops evidence-based explanation design patterns.

The study in [33] proposes integrating dataset documentation (following datasheet conventions) into requirement management systems to maintain ethical traceability. This enables linking requirements to data characteristics and limitations. Implementation requires tool extensions and process changes. Future work develops integrated documentation-management tools.

The study in [34] advocates for integrating model cards and similar documentation into requirement management systems, creating bidirectional traceability between requirements and model characteristics. This supports impact analysis when models change. Technical implementation presents challenges of integration. Future work develops standardized integration APIs.

The study in [35] proposes runtime monitoring systems to continuously assess ethical compliance of deployed AI systems through metrics collection and analysis. The approach enables proactive intervention but requires careful metric selection and threshold setting. Future work develops adaptive monitoring frameworks.

The study in [36] presents frameworks for post-deployment auditing of AI systems, including periodic reviews, incident analysis, and impact assessments. Effective auditing requires access to system data and clear accountability structures. The research provides audit templates but notes implementation variability. Future work develops sector-specific audit protocols.

The study in [37] reports low adoption rates of ethical RE practices in industry despite growing awareness, identifying organizational, technical, and cultural barriers. Surveys of 200 organizations reveal implementation gaps between policy and practice. The research quantifies adoption challenges but offers limited intervention strategies. Future work develops adoption acceleration programs.

The study in [38] identifies complexity and lack of automation as major barriers to adopting ontology-based and assurance-case ethical frameworks. Simplification and tool support are essential for practical adoption. The research analyzes complexity sources but provides minimal simplification guidance. Future work develops simplified framework variants.

The study in [39] documents strong industry demand for practical, tool-supported approaches to ethical RE that bridge theory and practice. Industry surveys prioritize usability, integration, and clear ROI over theoretical completeness. The research identifies requirements for practical tools but doesn't develop solutions. Future work focuses on industry-aligned tool development.

The study in [40] analyzes EU GDPR Articles 13-15 requirements for transparency and the "right to explanation" in automated decision-making systems. The research interprets legal requirements in technical terms and identifies implementation approaches. Legal interpretations continue to evolve. Future work monitors regulatory developments and develops compliance frameworks.

Datasets Description

Dataset 1: Adult Income Dataset from UCI

This data consists of the demographic data of 1994 U.S census age, gender, income, occupation, and education. The study will aim to examine the prejudice and integrity of income forecasting models. The racial or gender prejudice is often researched with the help of the dataset, which is why it is an attractive fact that supports the validation of ethical requirements.

Dataset 2: COMPAS Recidivism Dataset

The COMPAS algorithm process is based on the behavioral, demographic, and criminal history of the dataset to predict a potential risk of a person recidivating. The objective of the study is to

demonstrate how ethical conceptual principles such as fairness and transparency can be implemented in AI-based decision-making systems by experimenting with bias and accountability indicators within the framework of the RE.

Dataset 3: German Credit Dataset

The information about the job status, credit history, and financial situation of people is used in this dataset that is utilized to estimate credit risk. It will help in evaluating the requirements of explainability, privacy and fairness in the financial decision making systems. It is possible to substantiate the moral standards related to responsible lending and equal treatment due to the dataset.

Dataset 4: Bank Marketing Dataset (UCI Repository)

This dataset gives the demographics of the clients and the outcome of the marketing campaign of a Portuguese banking institution. The purpose of the study is to show how a clear decision logic could make the consumer more trusting of AI-based marketing by testing explainability and accountability criteria..

Table 1: Datasets Description

Dataset Name	Size	Features	Protected Attributes	Target Label	Purpose in Research
Adult Income Dataset (UCI Census Income)	48,842 rows	age, workclass, education, marital-status, occupation, relationship, race, sex, hours-per-week, native-country	Race, Gender	income (>50K or ≤50K)	• Test gender & racial bias in income prediction using fairness metrics. • Validate fairness, transparency requirements using SHAP, LIME, AIF360.
COMPAS Recidivism Dataset	7,000+ rows	age, age_cat, priors_count, juv_fel_count, juv_misd_count, charge_degree,	Race, Gender	risk score (low/medium/high)	• Known biased dataset → ideal to demonstrate fairness violations. • Used to evaluate fairness metrics and

		days_b_screening_arrest			mitigation in your RE framework.
German Credit Dataset (UCI)	1,000 rows	credit history, purpose, employment status, personal-status, housing, savings, age	Age, Gender (encoded inside personal-status)	credit risk (good/bad)	• Test discrimination in credit scoring. • Validate fairness and accountability requirements. • Good for SHAP explainability demonstrations.
Bank Marketing Dataset (UCI)	45,211 rows	age, job, marital, education, balance, housing, loan, campaign, duration	Age (Only one)	subscribed to term deposit (yes/no)	• Used to test fairness when dataset has limited protected attributes. • Good for demonstrating transparency and accountability (SHAP/LIME).

Table 2: Tentative Timetable

Process	Sep, Oct	Nov, Dec	Jan	Feb, Mar	Apr, May	June
Planning	Topic selection + Intro					
Documentation		Literature Review				
Implementation			Development			
Development				Coding		

Debugging					Debugging + Thesis	
Testing						Thesis Review

Potential Impact and Contribution to Scientific Community and Society

This study will be valuable to the scientific community because it provides a systematic structure that considers ethical implications in software requirements engineering of AI systems. It will present a systematic approach towards identifying, documenting and attesting moral norms such as accountability, openness and equity. Also, the research will contribute to the development of an online validation tool that will enable developers and researchers to check the level of ethical compliance when developing systems.

The research aims at improving the development of more responsible and trustworthy AI systems by society. The proposed solution will help to reduce such issues as prejudice, discrimination, and the lack of transparency in AI decision-making by involving ethical concerns in its initial stages of development. Finally, the work will aid in establishing AI systems that do not undermine human values, increase the confidence of people and advance technology in a sustainable and fair way.

One Most Relevant Industry

Steller Stack

One Targeted Product (Name, Type, Description)

Product Name: EthicalReq-AI

Type: Web-Based Requirements of Ethical Requirement Engineer and Requirement validation tool.

EthicalReq-AI web-based platform is designed to assist in the identification, documentation, and certification of the ethical requirements within artificial intelligence (AI) systems. In the platform, researchers and developers can be capable of capturing ethical concepts like accountability, explainability, openness, and fairness and associate them with pertinent models, datasets and metrics of assessment. It will also offer automatic validation tests all through the AI development lifecycle in order to assure that ethical standards will be attained at all times.

In addition, the site will have compliance reporting tools, ethical risk assessments tools and requirement traceability tools. All these features will empower the EthicalReq-AI to help enterprises create credible and ethical solutions with references to AI by bridging the gap between the theoretical ethical models and the use of AI.

Current Status of the Product

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Certificate of Originality

I __Sabahat Siddique__, Reg. No. __M24F0023AI003__, MS Scholar at School of Computing Sciences, certify that my MS research proposal with title “__Requirement

Engineering for AI Systems__” declare the originality of contents that has been verified through plagiarism software and is _____%.

Student Name: Sabahat Siddique

Reg. No: M24F0023AI003

Signature: _____

Date: _____

Supervisor

Name: _____

Signature: _____

Date: _____

Recommendation by the Research Supervisor

I have read the synopsis entitled

_____ and agree to supervise Mr./Ms. _____ for the partial requirements of the degree of MS in _____. The proposed project is academically, logistically, administratively, and financially feasible and all the required literature, equipment, laboratories, space and transport facilities are or will be available at PAF-IASST, Haripur or at _____.

The following facilities/finances will be required for this project:

Financial Requirements:

Estimated Cost: Rs. _____

These financial requirements will be fulfilled from _____ source(s).

Supervisor's Name: _____ Signature _____ Date _____

Signed by GEC Member

S. #	Name of Committee Member	Designation	Signature and Date
1			
2			
3			
4			

Recommendation by Departmental Graduate Committee (DGC)

Certified that the synopsis has been seen by members of DGC and considered it suitable for putting up to the Advanced Studies and Research Board of PAF-IAST (ASRB).

Graduate Coordinator: _____ Signature: _____ Date: _____

Convener DGC (Chairman/HoD): _____ Signature: _____ Date: _____

Dean, Faculty of

_____ Recommended for placement before ASRB.

_____ Not Approved on the basis of following reasons

Signature: _____

Date: _____

Secretary ASRB

_____ Approved from ASRB.

_____ Not Approved on the basis of the following reasons

Signature _____

Date _____